

Experiment 1 – Working with Python Packages – NumPy, SciPy, Scikit – Learn, Matplotlib

Aim

To explore the python data libraries to apply them in real datasets to perform Exploratory Data Analysis on these datasets using summary statistical data and visualisations such as box plots, scatter plots and correlation heatmaps.

Datasets

Iris Dataset: Consists of 150 rows of iris flowers with 4 features (sepal width, length, petal length, width) and a target column species. This is a classification problem.

Loan Approval Dataset: Dataset with numeric attributes (income, loan amount, asset values) and a target column loan_status. This is a binary classification problem

EDA Implementation:

A reusable `perform_eda()` function is created to perform EDA operations for any tabular dataset.

It

- Strips all whitespace from the column names
- Prints the dataset shape and drops any columns like id, index etc
- Fills missing values using the columnar median
- Uses a set of usual target words to find the target column and infers the type of problem
- Generates a multi graph – image (box plots, heatmaps and scatterplot) in a .eps file
- Prints per – feature statistics – mean, median, standard deviation, mix, max, quartiles, skew and kurtosis

Iris Dataset – EDA:

The dataset has no missing values and is perfectly balanced. Figure 1 shows the automated EDA function in use and the output that is obtained.

Figure 2 shows the combined view of the distributions and relationships across all columns.

Figure 3 shows a pairplot which visualizes the pairwise scatter relationships between every pair of features with hue representing different species.

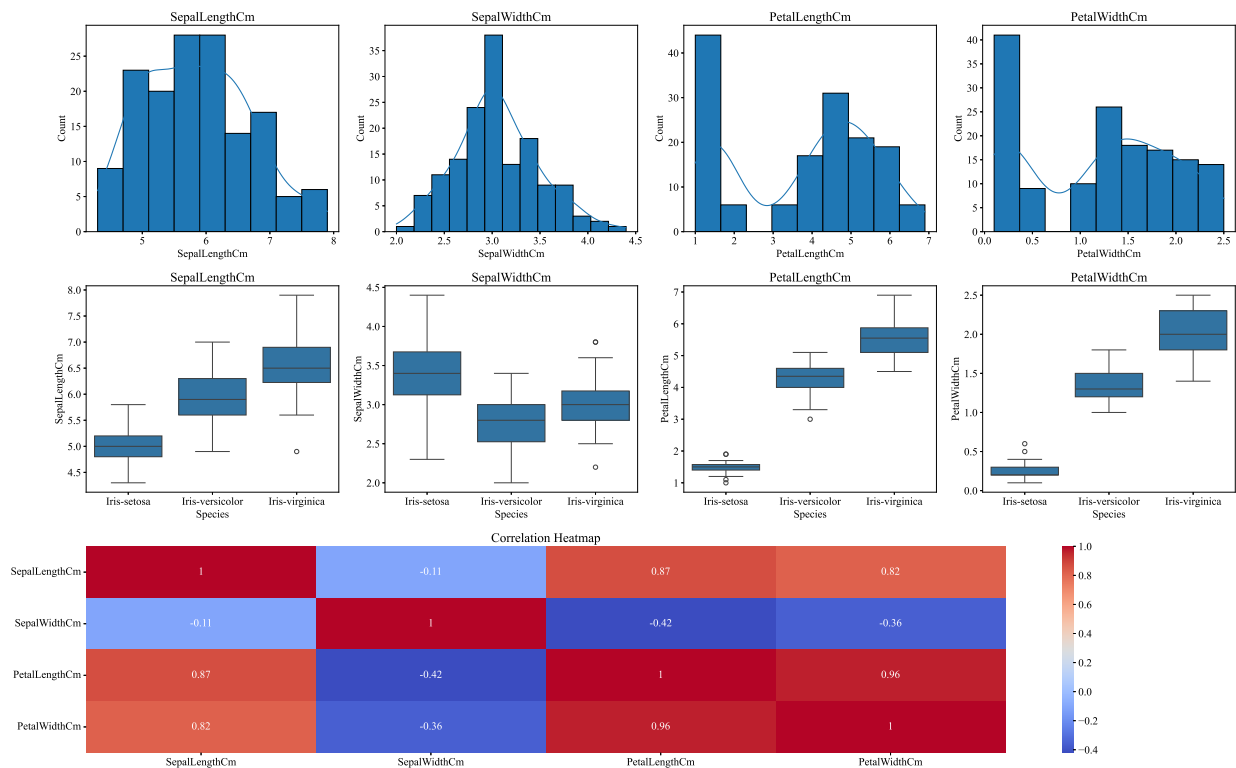


Figure 1: Automated EDA function in use and the output that is obtained (Iris dataset).

Loan Approval Dataset – EDA:

The loan approval dataset noisier than the iris dataset with the financial columns using different scales.

CIBIL score shows a different distribution between approved and rejected status making it the strongest predicting column

Income and loan amount are right skewed with high number of high – income applicants

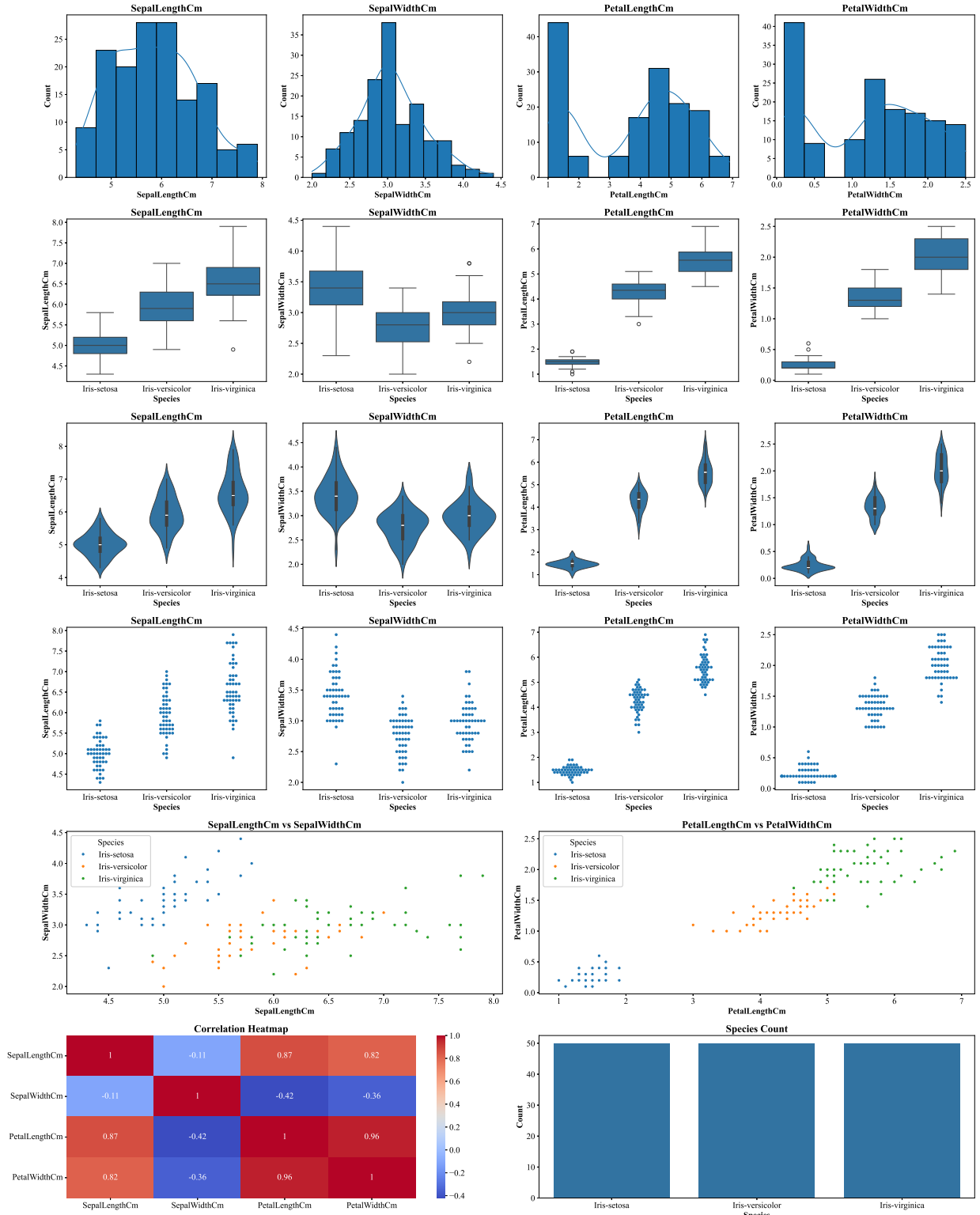


Figure 2: Combined view of the distributions and relationships across all columns (Iris dataset).

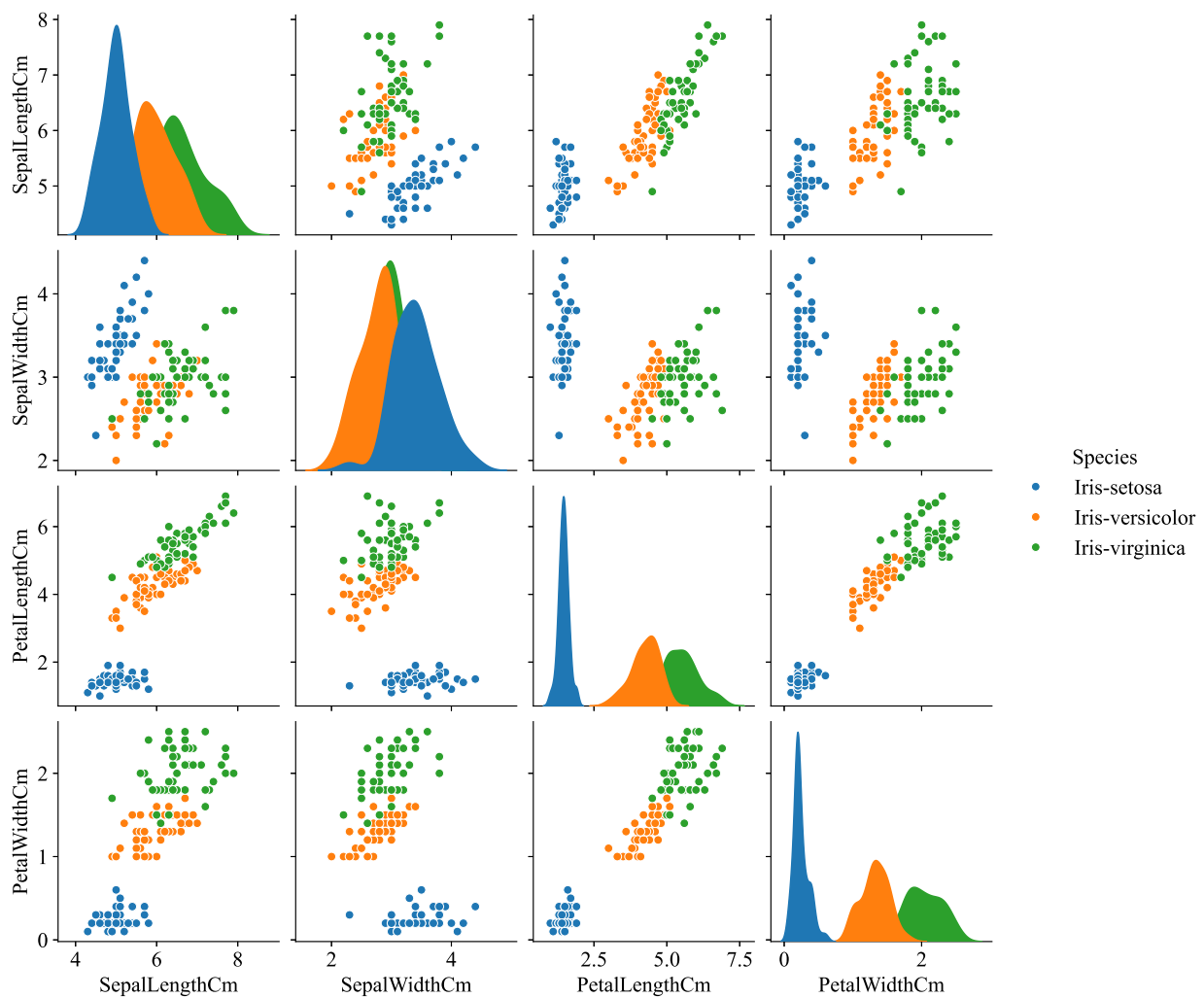


Figure 3: Pairplot visualizing the pairwise scatter relationships between every pair of features with hue representing different species.

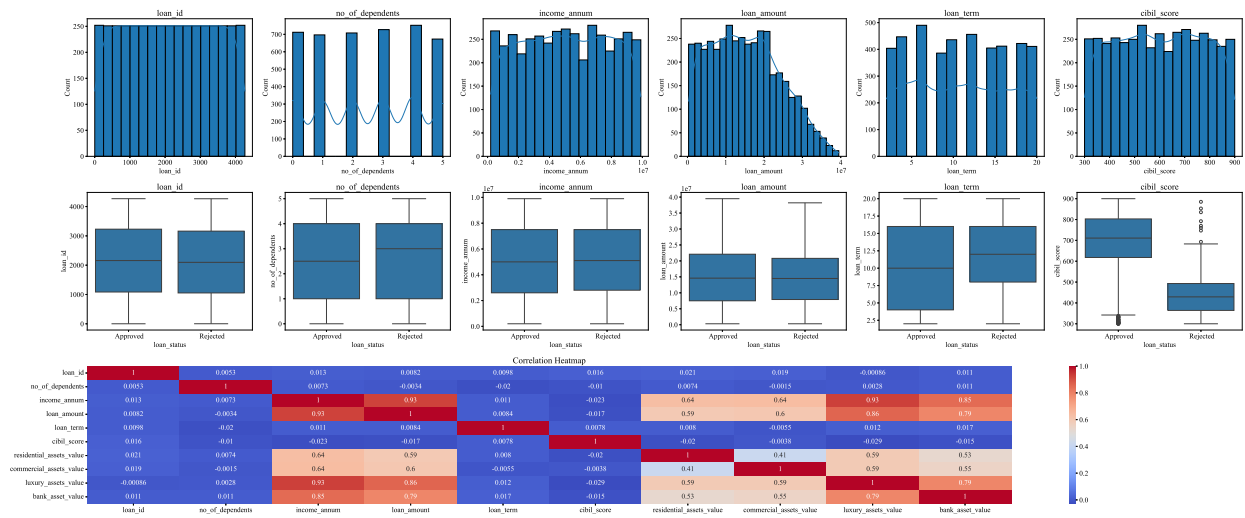


Figure 4: Automated EDA function output for the Loan Approval dataset.

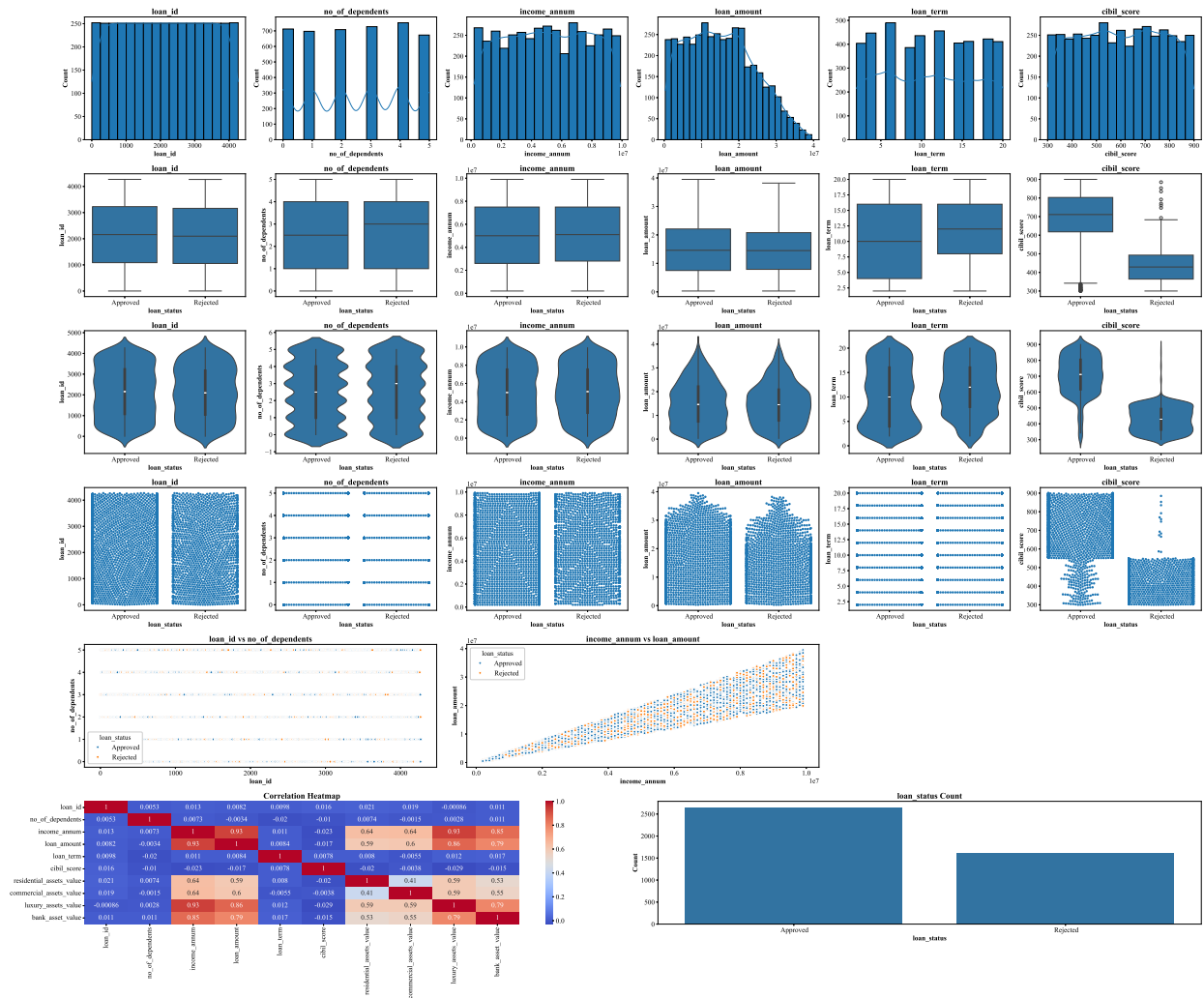


Figure 5: Combined view of the distributions and relationships across all columns (Loan Approval dataset).